

One-bath Open-chain Profile Diagnostic

Frozen numerical audit

September 2026

1 Boundary definition

The left endpoint carries the canonical BC1 bath,

$$b_{I,L} = 2\gamma [2T_L - (2MI_1 - I_1^2 + 2I_1I_2 \cos \delta_1)], \quad b_{\phi,L} = 2\gamma I_2 \sin \delta_1,$$

with $\sigma_{I,L} = 2\sqrt{2\gamma T_L I_1}$ and $\sigma_{\phi,L} = \sqrt{2\gamma T_L / I_1}$. At the free endpoint, $b_{I,R} = b_{\phi,R} = \sigma_{I,R} = \sigma_{\phi,R} = 0$ exactly; only Hamiltonian drift remains. Thus this is the genuinely open definition from `flux_V2.cpp`, not the non-open historical `main_fixed.cpp` cases.

The port deliberately retains the corrected-noise SIMD profile integrator so the action profiles can be compared directly with the existing 72-run matrix. It also retains single precision, adaptive Euler–Maruyama stepping, polynomial trigonometry, and positive-action projection. The saved bond-sine values are therefore archival only: the independent controlled-integrator audit showed that this scheme produces a spurious sine signal.

2 Stationarity result (reported first)

The open-chain stationarity gate does not pass for every condition. Failed conditions: T10_n100. Scaling and collapse numbers involving these conditions are reported descriptively and are not stationary-profile claims.

The gate compares the separate third and fourth measurement quarters. It requires total-action and free-end paired differences to contain zero and be at most 5 percent in relative magnitude; the interior sitewise median and maximum changes must be at most 5 and 10 percent, respectively.

T_L	n	ΔM	SE	rel.	CI0	ΔI_n	SE	rel.	CI0	bulk med.	bulk max	pass
T10	25	-0.09640	0.05392	0.0060	yes	-0.00600	0.00415	0.0099	yes	0.0069	0.0150	yes
T10	50	-0.05412	0.09503	0.0023	yes	-0.00768	0.00473	0.0190	yes	0.0091	0.0185	yes
T10	100	0.03707	0.09089	0.0011	yes	-0.00903	0.00435	0.0321	no	0.0131	0.0391	no
T6	25	-0.01126	0.03961	9.0526e-04	yes	-5.31530e-04	0.00343	0.0011	yes	0.0025	0.0075	yes
T6	50	-0.04962	0.07597	0.0028	yes	0.00103	0.00395	0.0032	yes	0.0101	0.0236	yes
T6	100	-0.03260	0.06563	0.0013	yes	0.00102	0.00346	0.0046	yes	0.0163	0.0371	yes

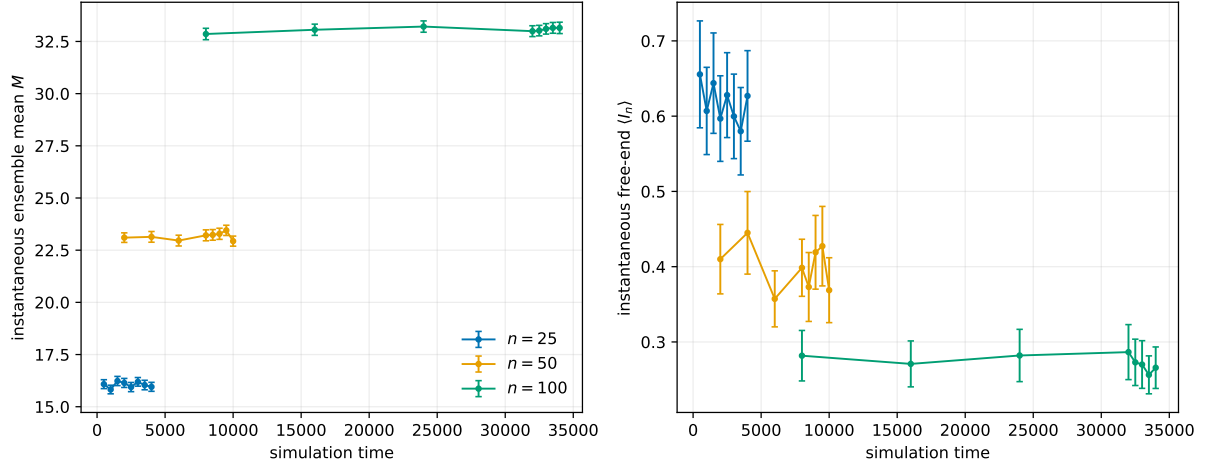


Figure 1: Stationarity diagnostics at left-bath temperature 10: instantaneous total action and free-end action.

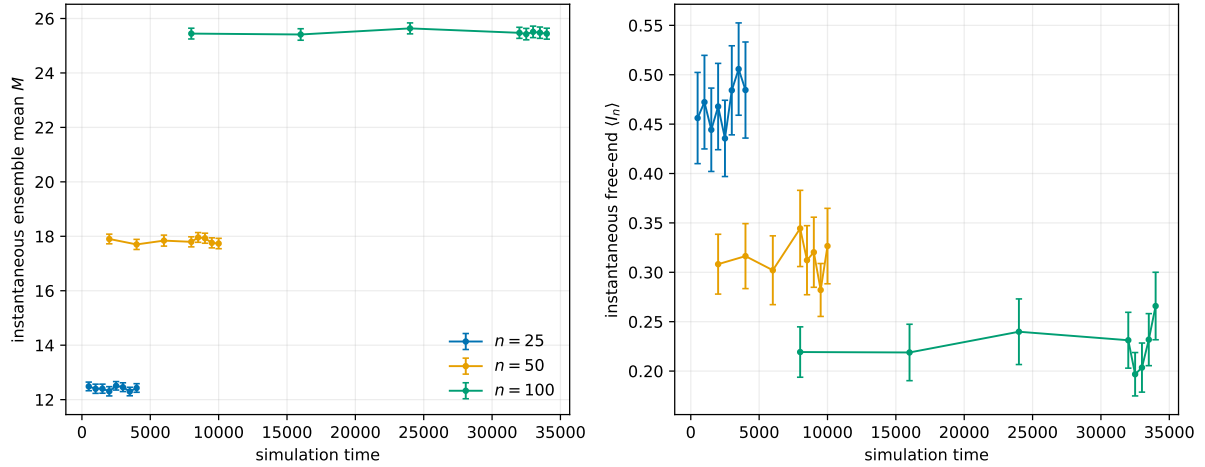


Figure 2: Stationarity diagnostics at left-bath temperature 6: instantaneous total action and free-end action.

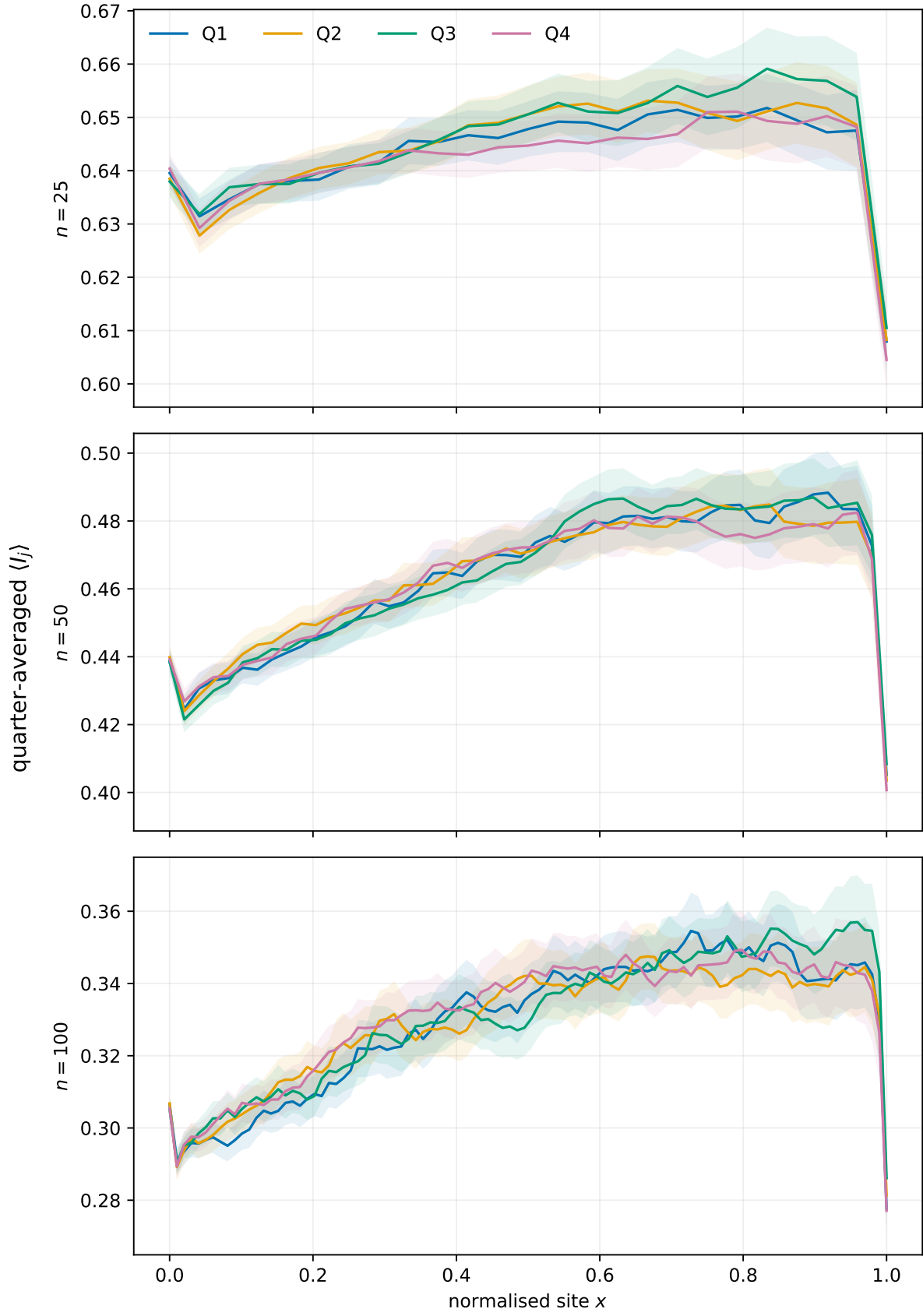


Figure 3: Separate quarter-averaged action profiles at left-bath temperature 10.

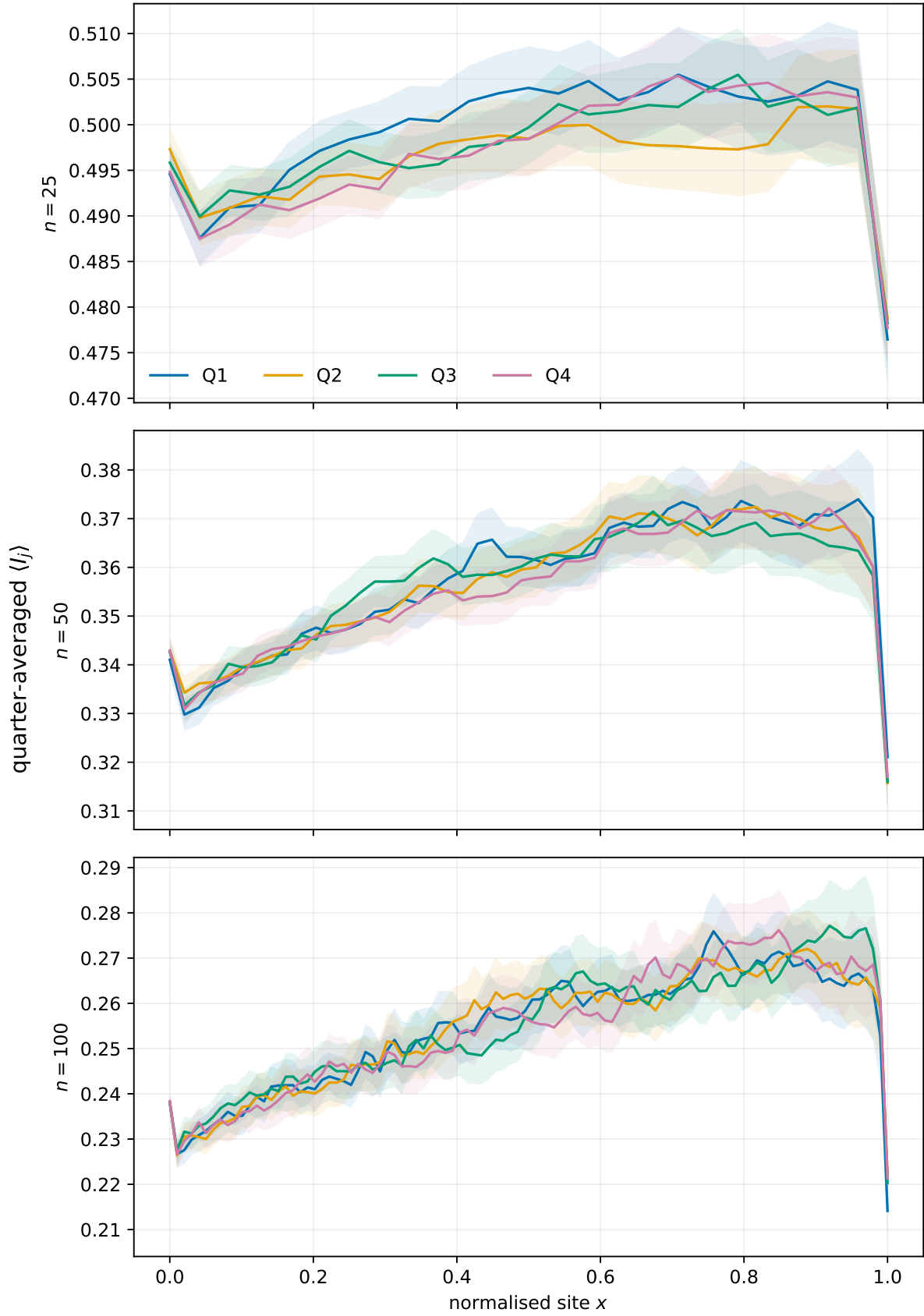


Figure 4: Separate quarter-averaged action profiles at left-bath temperature 6.

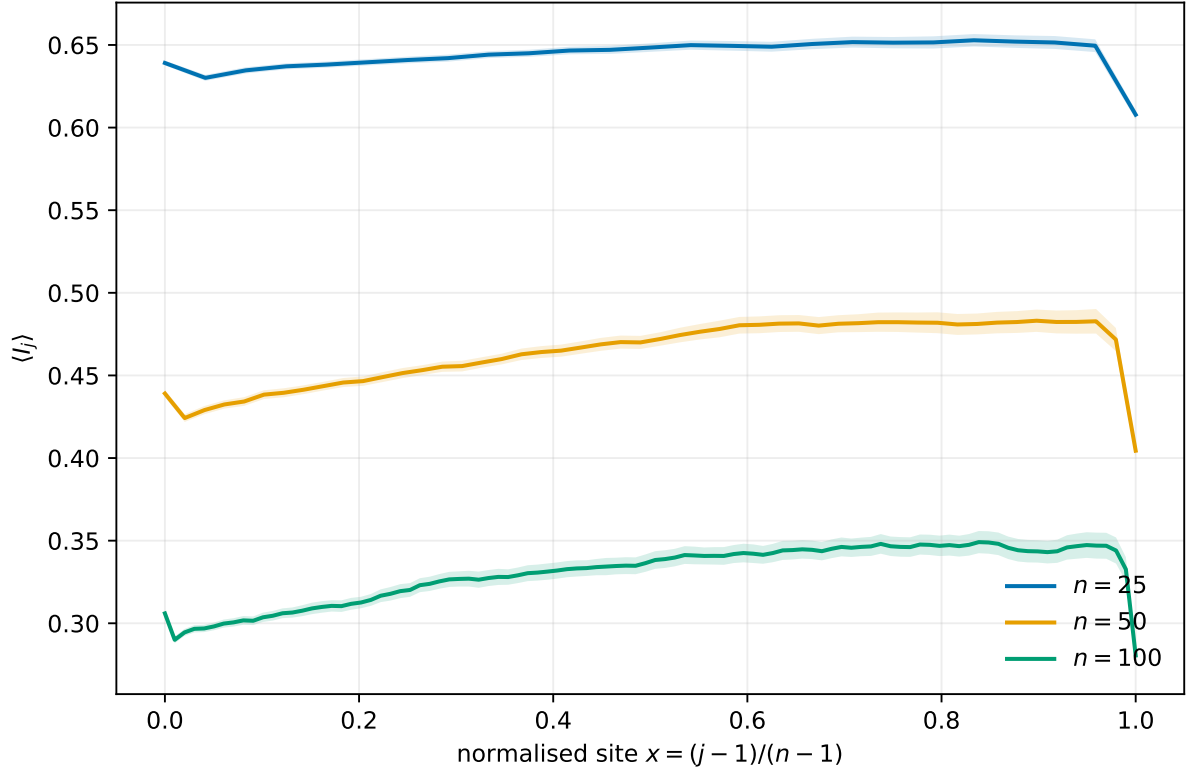


Figure 5: Raw open-chain action profiles at left-bath temperature 10; bands are pointwise 95 percent intervals.

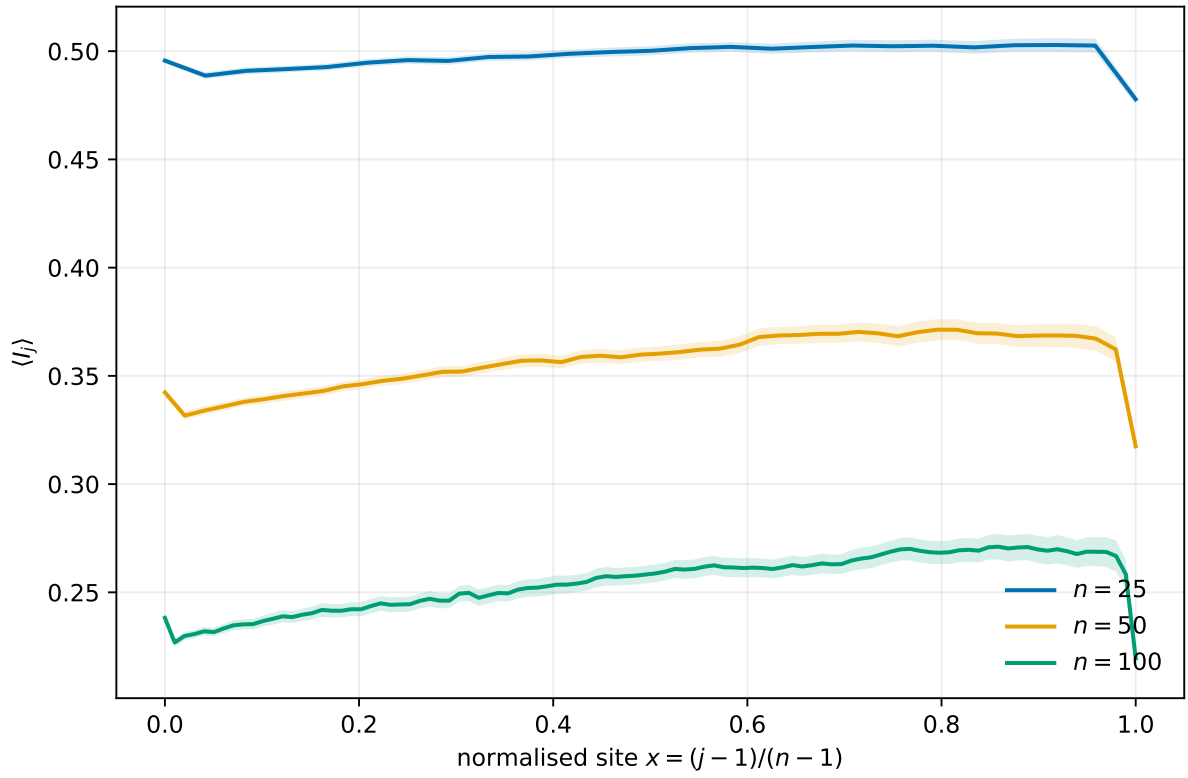


Figure 6: Raw open-chain action profiles at left-bath temperature 6; bands are pointwise 95 percent intervals.

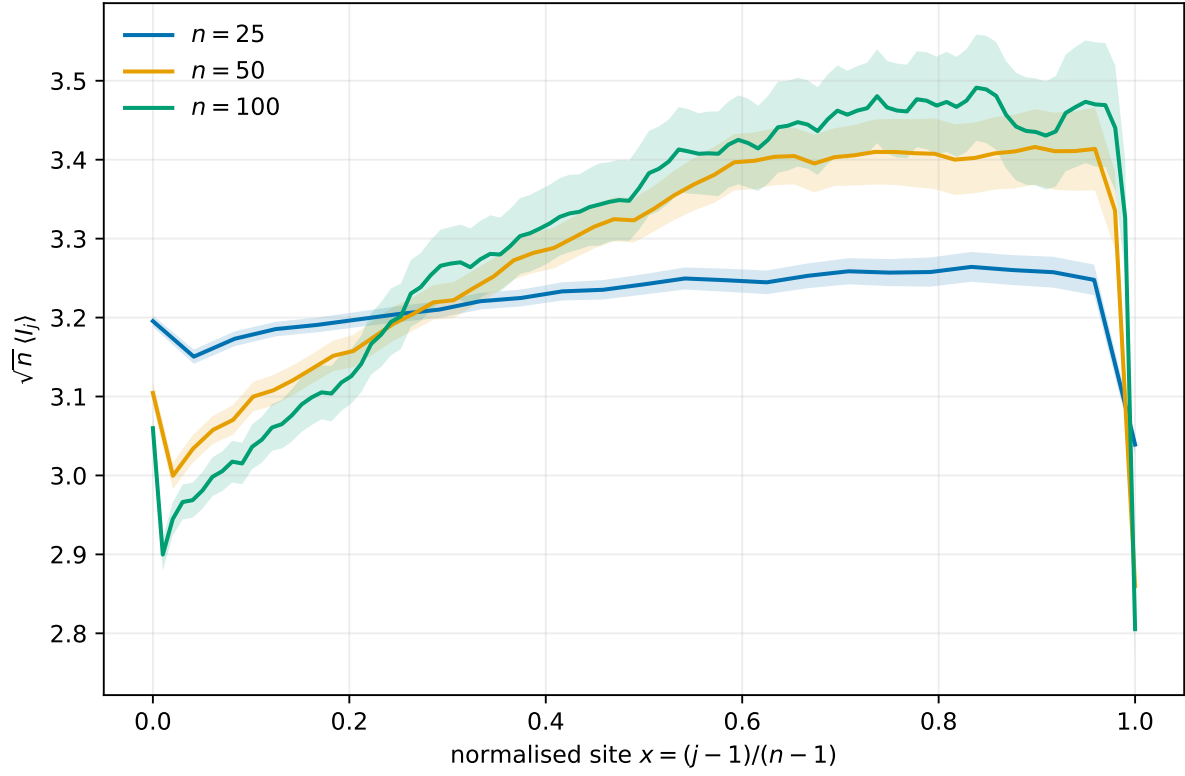


Figure 7: Fixed-exponent collapse test $\sqrt{n}\langle I_j \rangle$ at left-bath temperature 10.

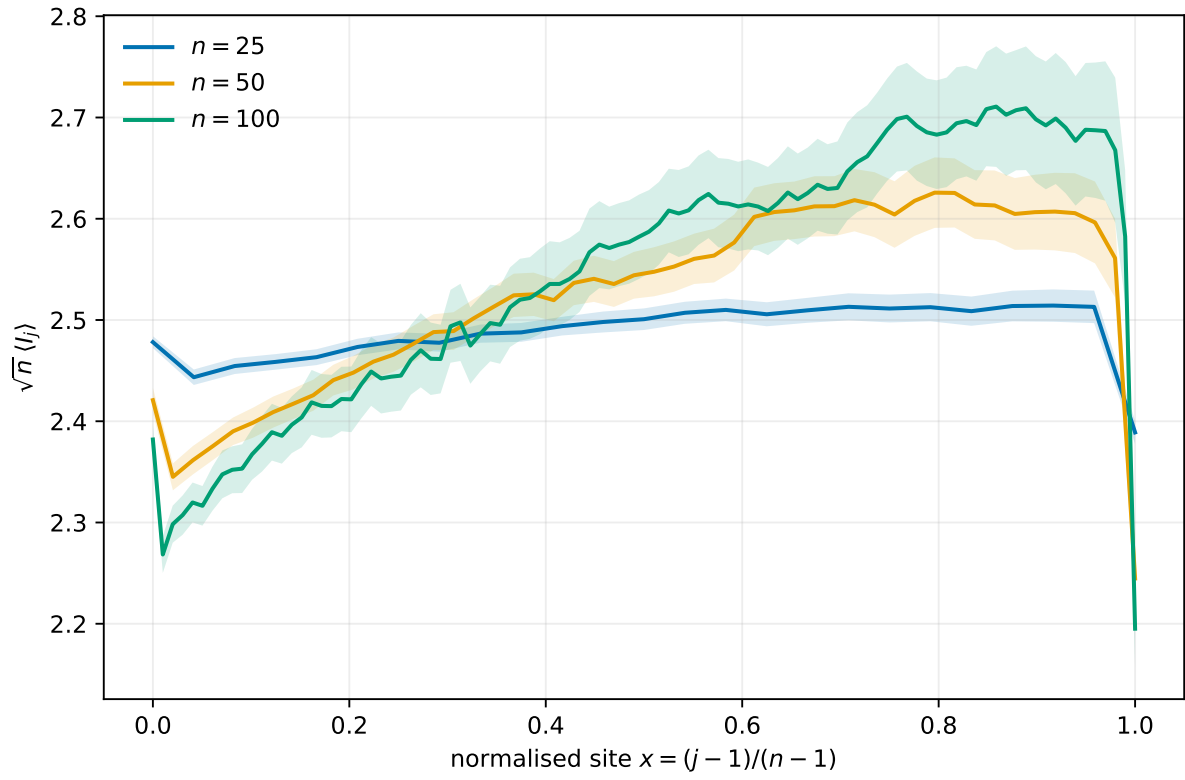


Figure 8: Fixed-exponent collapse test $\sqrt{n}\langle I_j \rangle$ at left-bath temperature 6.

3 Mid-chain and free-end scaling

T_L	metric	n	site	mean	SE	stationary
T10	free_end	25	25	0.60781	0.00139	yes
T10	free_end	50	50	0.40444	0.00207	yes
T10	free_end	100	100	0.28052	0.00223	no
T10	mid_chain	25	13	0.64842	0.00132	yes
T10	mid_chain	50	25	0.46997	0.00203	yes
T10	mid_chain	100	50	0.33639	0.00268	no
T6	free_end	25	25	0.47779	0.00118	yes
T6	free_end	50	50	0.31745	0.00165	yes
T6	free_end	100	100	0.21951	0.00198	yes
T6	mid_chain	25	13	0.50016	0.00109	yes
T6	mid_chain	50	25	0.35980	0.00167	yes
T6	mid_chain	100	50	0.25822	0.00213	yes

T_L	metric	exponent	OLS SE	95% CI ($df = 1$)	log-SSE	all stationary
T10	mid_chain	-0.47339	0.00523	[-0.53982, -0.40696]	2.62645e-05	no
T10	free_end	-0.55776	0.01729	[-0.77748, -0.33804]	2.87333e-04	no
T6	mid_chain	-0.47689	9.86565e-04	[-0.48942, -0.46435]	9.35260e-07	yes
T6	free_end	-0.56105	0.01663	[-0.77239, -0.34971]	2.65841e-04	yes

Each exponent uses only three chain lengths and therefore has one residual degree of freedom. No exponent is adjusted to improve collapse.

4 Fixed $\sqrt{n}$ collapse

Profiles are linearly interpolated to the predeclared 101-point grid $x = 0, 0.01, \dots, 1$. Relative spread is $(\max_n y - \min_n y) / \text{mean}_n y$ and is reported on the full grid and fixed interior $0.10 \leq x \leq 0.90$.

T_L	scale	region	grid points	median spread	maximum spread
T10	raw	full	101	0.6449	0.7657
T10	raw	interior_0p10_0p90	81	0.6406	0.7237
T10	sqrt_n	full	101	0.0502	0.0930
T10	sqrt_n	interior_0p10_0p90	81	0.0439	0.0671
T6	raw	full	101	0.6496	0.7636
T6	raw	interior_0p10_0p90	81	0.6481	0.7160
T6	sqrt_n	full	101	0.0411	0.0852
T6	sqrt_n	interior_0p10_0p90	81	0.0349	0.0758

5 Penetration depth

The prospectively fixed threshold is 50 percent of the driven-end action. The first raw crossing and its linearly interpolated location are reported without tail extrapolation.

T_L	n	I_1	threshold	reached	first site	interpolated site	site/ n	stationary
T10	25	0.63910	0.31955	no	—	—	—	yes
T10	50	0.43905	0.21952	no	—	—	—	yes
T10	100	0.30599	0.15300	no	—	—	—	no
T6	25	0.49563	0.24782	no	—	—	—	yes
T6	50	0.34236	0.17118	no	—	—	—	yes
T6	100	0.23820	0.11910	no	—	—	—	yes

6 Run-level integrity and provenance

T_L	n	seed	threads	valid	nonfinite	projections	seconds
6	25	2026091425	2	256	0	1798104	181.94
6	25	2026091426	2	256	0	1796575	181.89
6	50	2026091450	2	256	0	6451523	820.35
6	50	2026091451	2	256	0	6462082	795.44
6	100	2026091500	2	256	0	31351515	5089.14
6	100	2026091501	2	256	0	31304163	4752.23
10	25	2026091225	2	256	0	2327157	193.93
10	25	2026091226	2	256	0	2328091	197.19
10	50	2026091250	2	256	0	8403924	780.44
10	50	2026091251	2	256	0	8414954	778.19
10	100	2026091300	2	256	0	40486534	5107.15
10	100	2026091301	2	256	0	40350081	4975.42

Parent profile-matrix source SHA-256: 40d13c7547d9c147ca8241ed701e1a1e85540fc0c21db48c4a63a943eccfe3ec.

Remote freeze commit: f4b74a06e658478e78b130975b34af20da9d9398.

Open-chain source SHA-256: dfee7ed0b416a30f2b27d919c524dfaf37e91ca44312ffbb9d68b7fc1cd36110.

Binary SHA-256: 09d801e71b7f3bd13cea56aba3d432b471dbf16a726a9866894148a91397575d.

Analysis SHA-256: 1fd79a163cfccc518cdcf95064dc994aad32f0854071692a1d00c29fbf69db2.

Exact commands, seeds, runtimes, complete raw files, merged profiles, separate quarter profiles, snapshot trajectories, spread curves, and all SHA-256 values accompany this report.